HealthyLife Tracker: A Web-Based BMI, Heart Rate, and Workout Logging System

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I. INTRODUCTION

A. Project Overview

In the domain of personal health, consistency is key to understanding one's physical well-being. "HealthyLife Tracker" is a web-based application designed to serve as a central repository for vital health metrics. While many individuals engage in physical activities, they often lack a unified system to store and view their physiological data alongside their workout history. This system fills that void by providing a dedicated digital platform where users can instantly compute their Body Mass Index (BMI), store Heart Rate (BPM) readings, and archive their Workout Logs. By consolidating these three critical components into a single interface, the system transforms raw health data into organized, accessible information.

B. Problem Statement

Despite the importance of tracking physical health, many individuals operate without a clear historical record of their progress. There is often a disconnect between knowing one's current weight or heart rate and understanding how these metrics change over time. Without a dedicated system to log and visualize this data, users miss valuable insights into their health trends. Furthermore, existing tools may be overly complex or fragmented, failing to provide a straightforward correlation between a user's BMI status, their pulse rate, and their daily exercise activities in a single view. This lack of data consolidation makes it difficult for users to maintain an accurate, long-term perspective on their physical fitness.

C. System Objectives

General Objective: The primary goal of the project is to develop a web-based "HealthyLife Tracker" that acts as a centralized database for calculating, recording, and visualizing user health metrics and workout history.

Specific Objectives:

1. To develop a system that accurately computes and categorizes Body Mass Index (BMI) based on user inputs.
2. To create a digital logging module that allows users to input and store Resting Heart Rate (BPM) and detailed Workout Logs (type, duration, date).
3. To implement data visualization tools that convert stored data into graphs, allowing users to visually track changes in their pulse and workout frequency.
4. To establish a secure database that ensures user health records are stored systematically for future retrieval.
5. To provide a unified dashboard that displays BMI status, heart rate history, and exercise logs simultaneously for better health awareness.

II. FEATURES OF THE SYSTEM

This section outlines the core capabilities of the application, designed to provide a centralized platform for health monitoring with distinct interfaces for general users and system administrators.

1. Secure User Authentication To ensure data privacy and system security, the application includes a robust login and registration module.

- **User Registration:** New users can create an account by providing necessary credentials, which creates a secure, personalized space for their health records.
- **Role-Based Login:** The login interface verifies user credentials and automatically directs them to the appropriate interface granting Administrators access to system oversight tools while restricting Standard Users to their personal health dashboards.
- **Session Management:** Ensures that a user's health data remains private and is only accessible while they are actively logged in.

2. ROLE-BASED DASHBOARDS (User & Admin) The system utilizes a secure login mechanism to direct users to their specific dashboard based on their access privileges.

1. Personalized User Dashboard: Designed for individual use, this interface serves as the personal command center for the user. It displays:

- A summary of the user's latest BMI, Heart Rate, and Workouts.
- Weekly Goal
- Recent Activity
- Daily Tip

2. Administrative Dashboard: Designed for system oversight, this interface allows the administrator to manage the application. Capabilities include:

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- **Global System Action:** The ability to reset all the data except on the dummy accounts.
- **User Management:** The ability to view the list of registered users within the system.
- **Delete User:** The ability to delete a specific user that is selected.
- **Database Oversight:** Ensuring the integrity of the records stored within the system.

3. HEALTH TOOLS

- **Automated Biometric Calculation (BMI)** The system features a built-in calculator that processes user input for height (cm) and weight (kg). It automatically computes the Body Mass Index (BMI) and provides an immediate classification of the user's weight status. To facilitate health awareness, it categorizes results into standard health zones:
    - Underweight
    - Normal Weight
    - Overweight
    - Obesity
- **Heart Rate (BPM) Data Entry & Monitoring** The application includes a module for recording the user's Resting Heart Rate (BPM). This feature allows users to digitally store their pulse rate readings, creating a vital sign record that helps users keep track of their cardiovascular baseline over time.

4. EXERCISE LOG

- **Digital Workout Journaling**
  - **Log & List**
    - **Log new workout -** A dedicated data entry interface that enables users to manually record specific exercise details, including the activity type, duration, date, and perceived intensity
    - **Weekly Trends -** A dynamic visualization tool that tracks physical activity volume over the past seven days, highlighting user consistency and progress toward weekly fitness targets.
    - **Activity Analysis -** A statistical breakdown that categorizes workout data by type providing users with insights into their training distribution and preferences.
  - **Calendar View**
    - **Workout Summary -** An aggregated overview that calculates key performance metrics, such as total accumulated exercise minutes and average workout duration, and calorie to provide a quick snapshot of overall physical effort

4. HISTORY AND GRAPHS

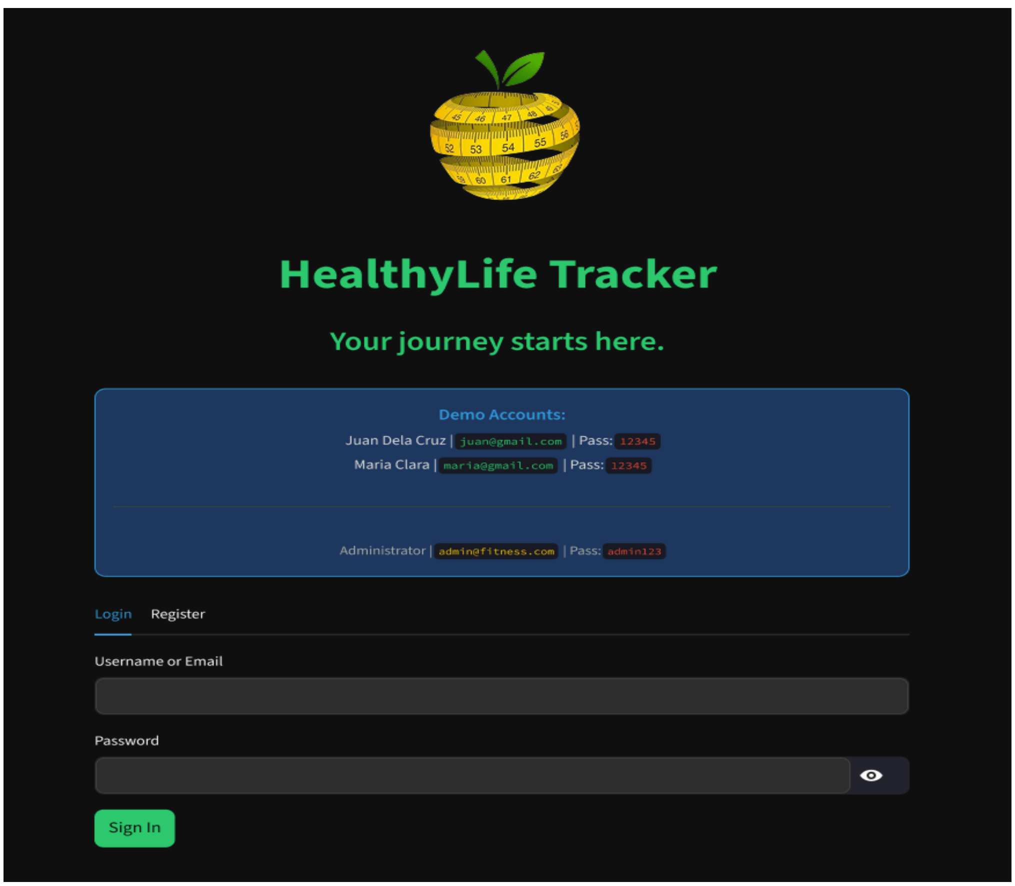
Integrated Data Visualization The system transforms raw user data into visual insights, moving beyond simple text-based logs.

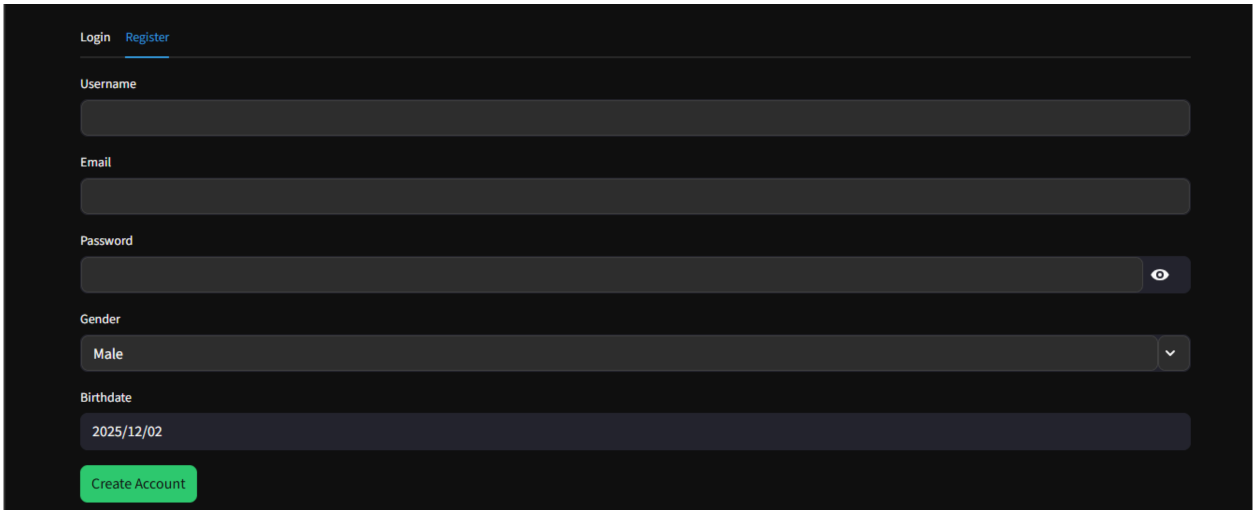
- **Medical Trends**

- **BMI History:** Visually displays the user's current BMI status in relation to healthy ranges and a detailed history where in the date it was recorded, bmi score, and its category.
- **Pulse History:** Visually displays the user's current Pulse status in relation to healthy ranges and a detailed history where in the date it was recorded, heart rate, and its status.

III. SCREENSHOT OF THE SYSTEM

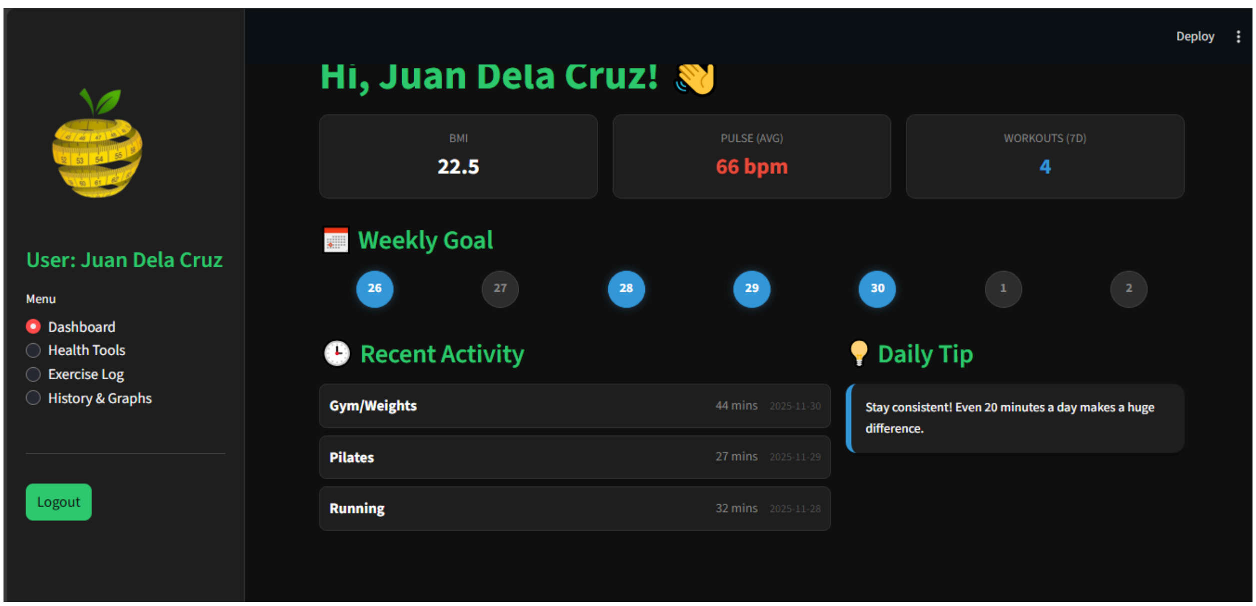
A. Secure User Authentication



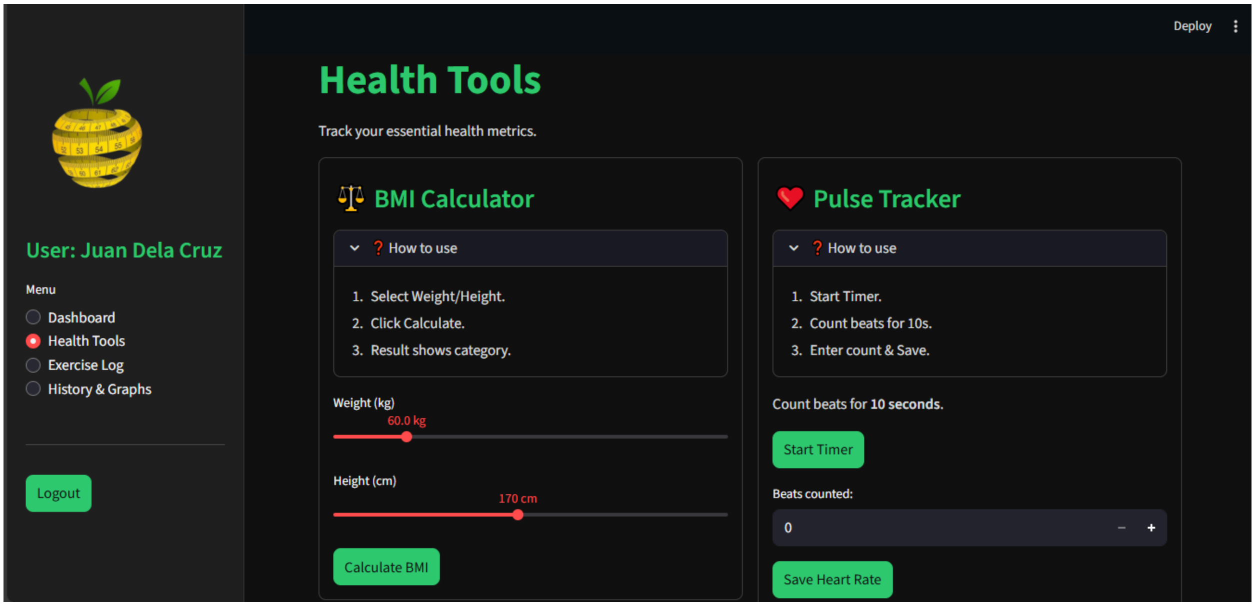


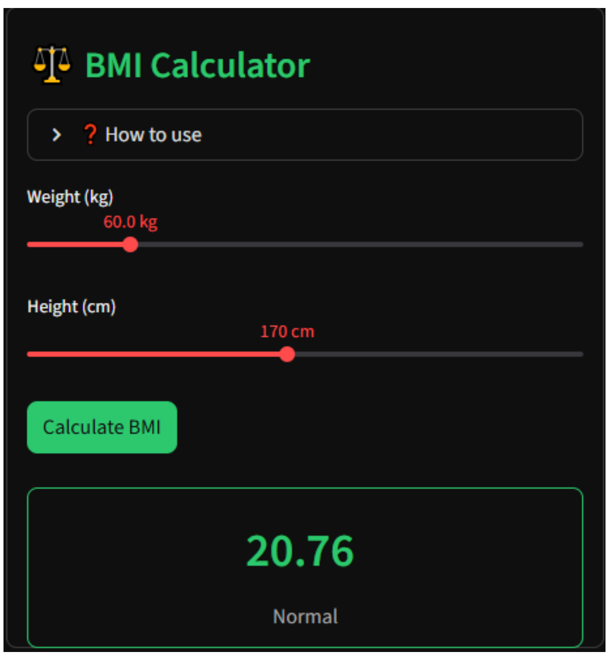
=== USER ACCOUNT ===

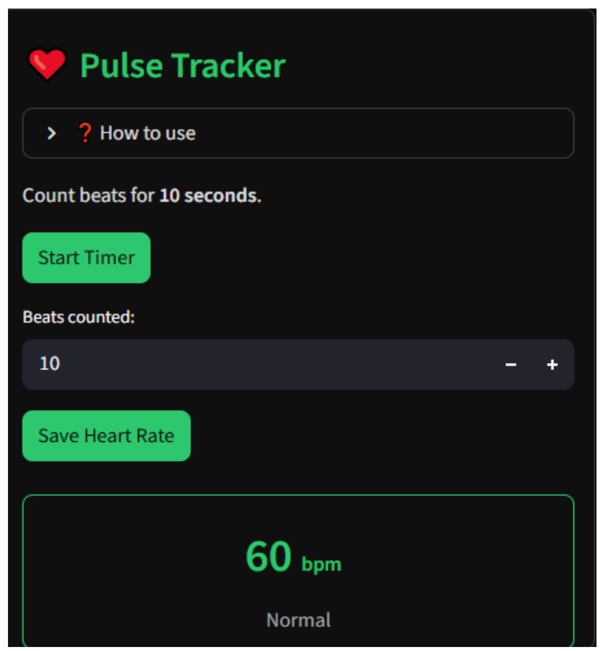
A. USER DASHBOARD



B. HEALTH TOOLS



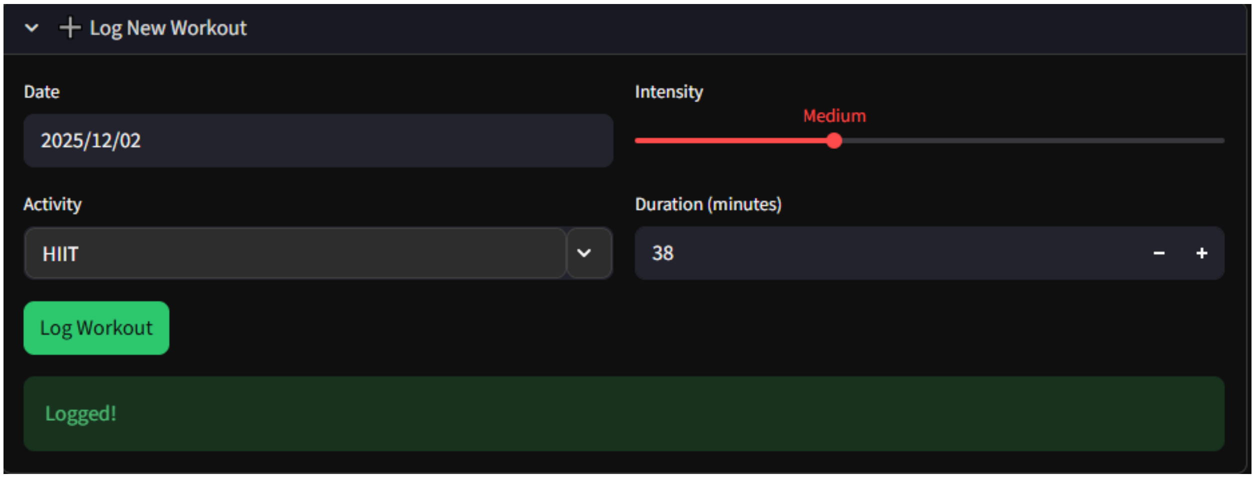




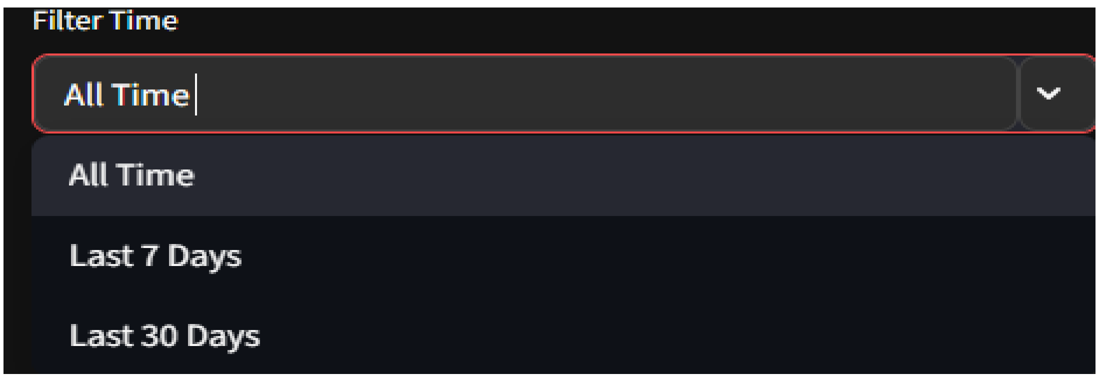
C. EXERCISE LOG

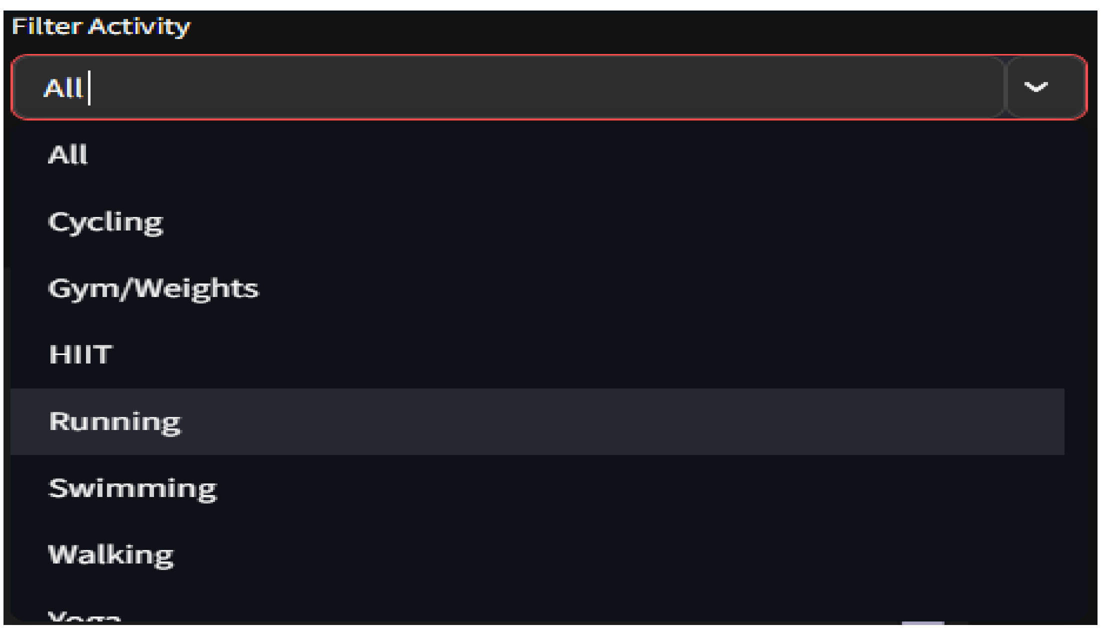
a. Log & List

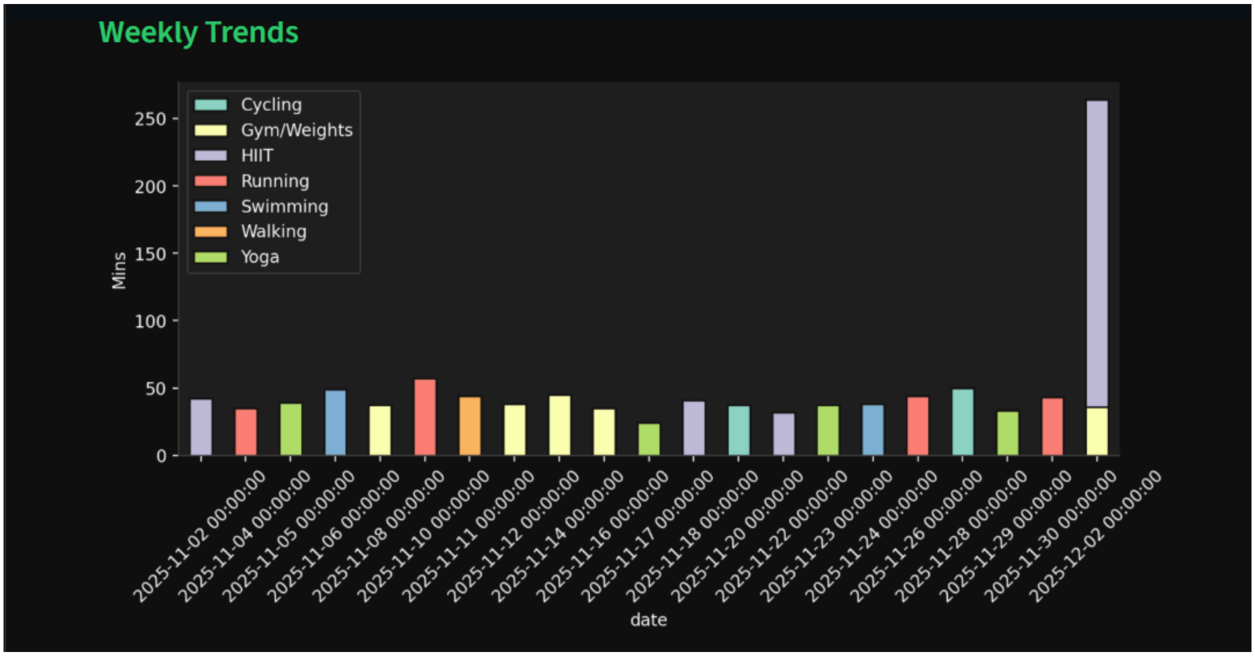
i. Log new Workout



ii. **Weekly Trends**







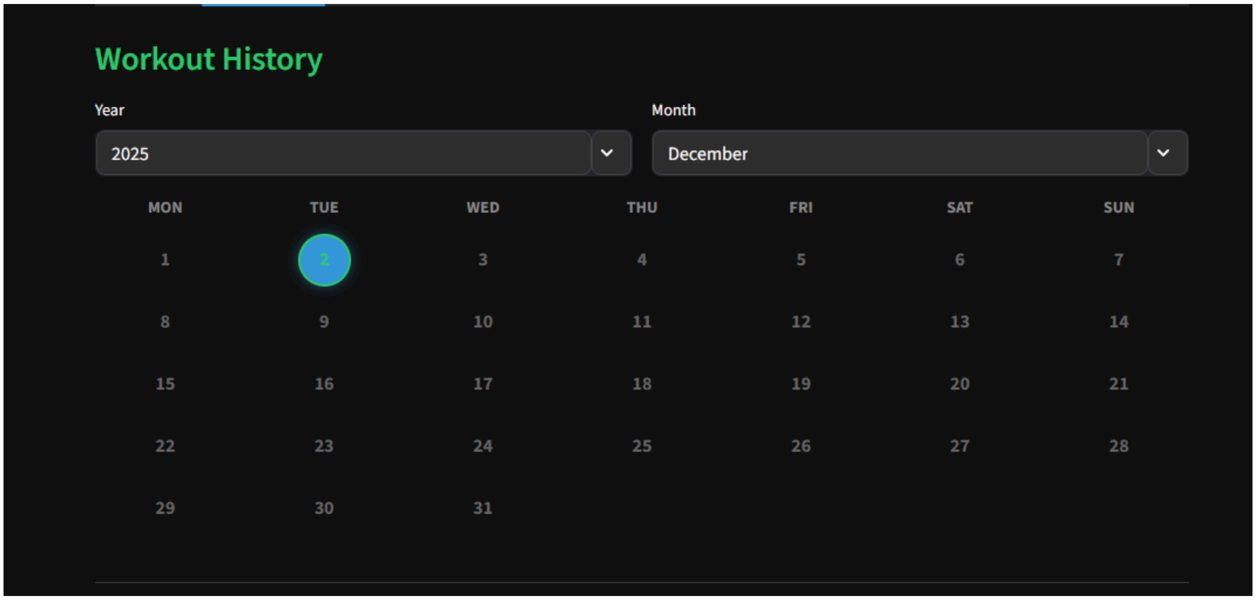
iii. Activity Analysis

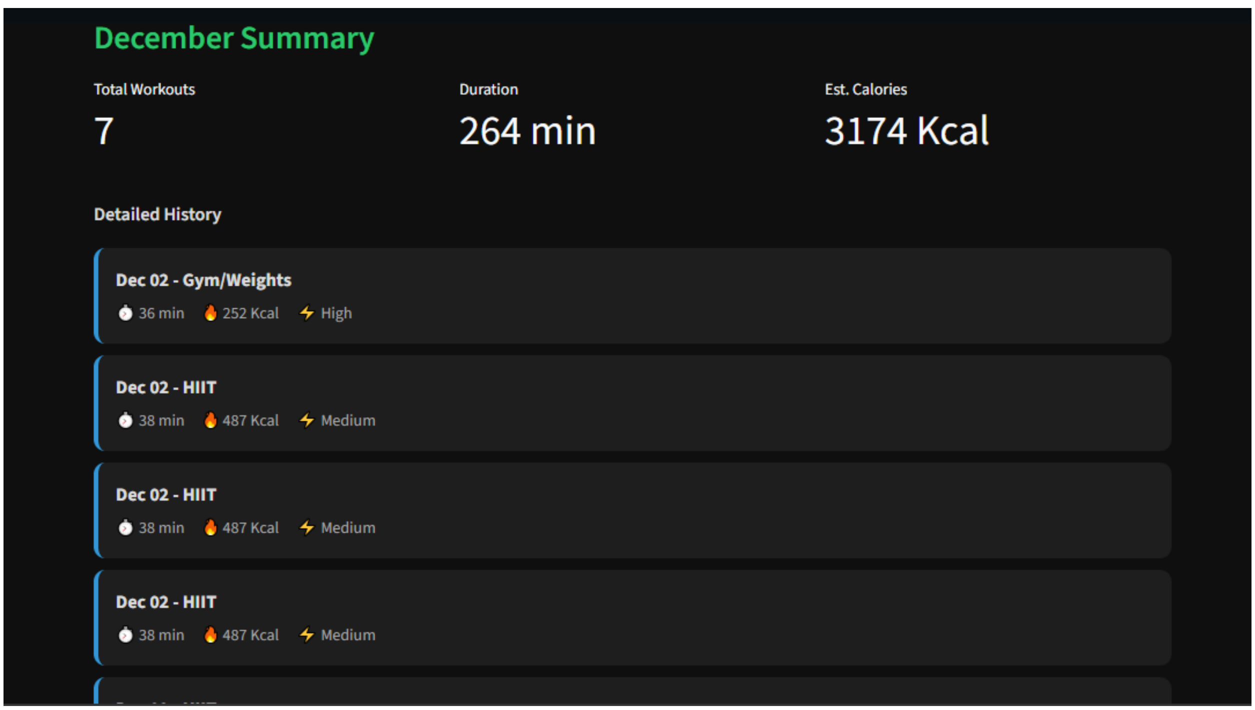
Activity Analysis

	date	activity	intensity	↑ duration
16	Nov 17, 2025	Yoga	Low	24
13	Nov 22, 2025	HIIT	High	32
8	Nov 29, 2025	Yoga	Low	33
17	Nov 16, 2025	Gym/Weights	High	35
25	Nov 04, 2025	Running	High	35
0	Dec 02, 2025	Gym/Weights	High	36
12	Nov 23, 2025	Yoga	Low	37
14	Nov 20, 2025	Cycling	High	37
22	Nov 08, 2025	Gym/Weights	High	37
1	Dec 02, 2025	HIIT	Medium	38

b. Calendar View

i. Workout History



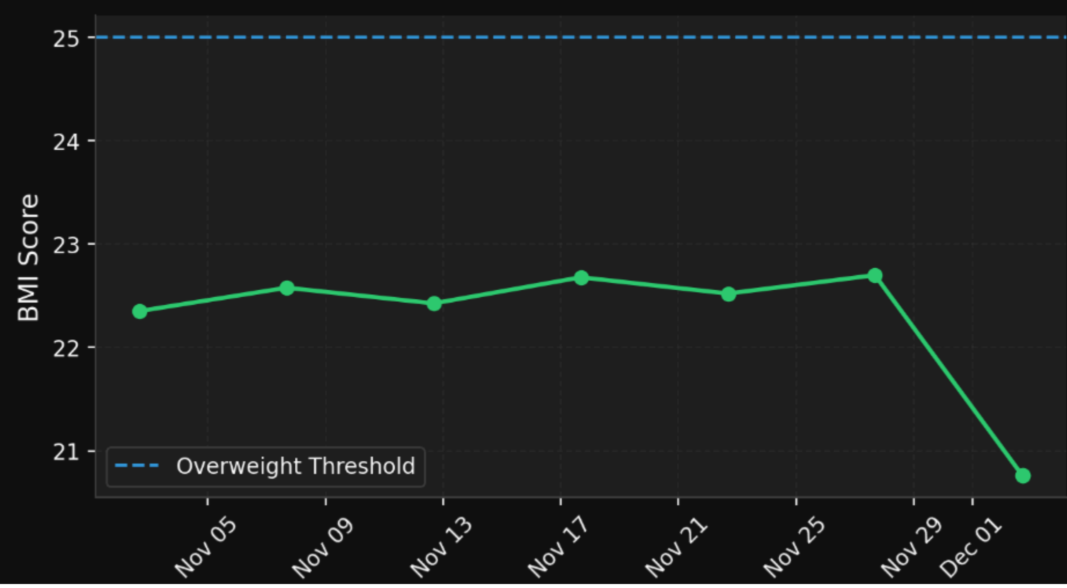


D. HISTORY & GRAPHS

a. Medical Trends

i. BMI History

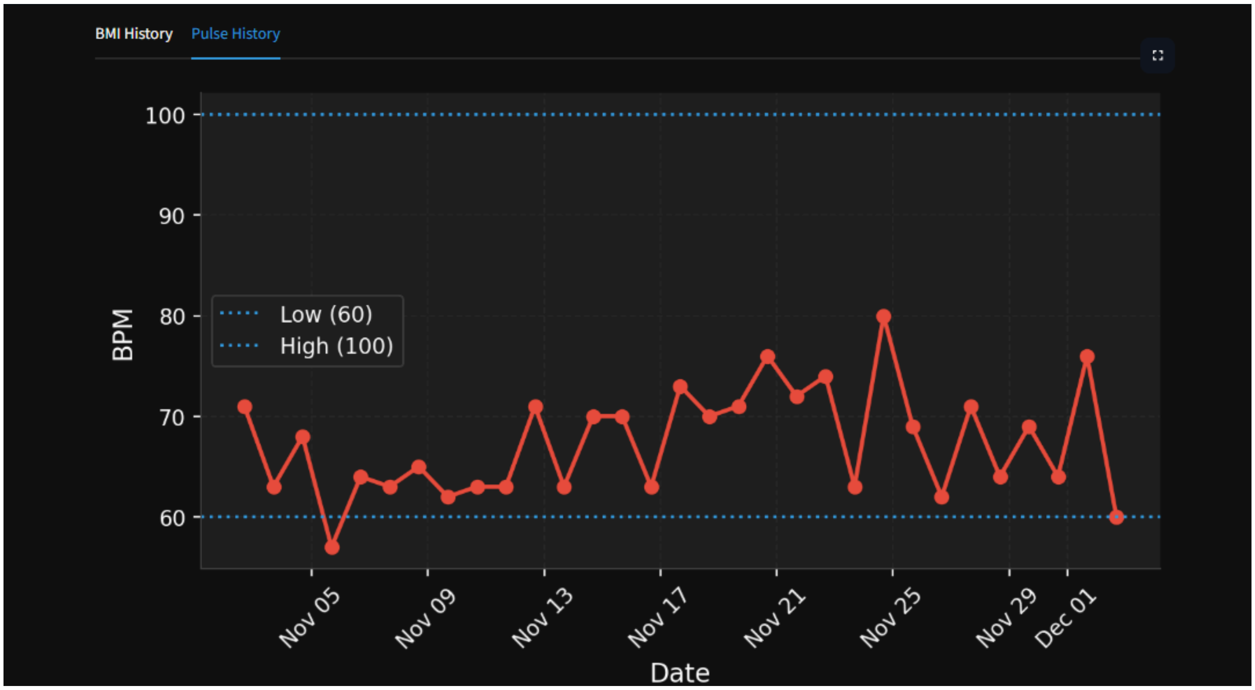
BMI History Pulse History



📊 Detailed History

	Date Recorded	BMI Score	Category
0	Dec 02, 2025 04:39 PM	20.76	Normal
1	Dec 02, 2025 04:39 PM	20.76	Normal
2	Dec 02, 2025 04:39 PM	20.76	Normal
3	Nov 27, 2025 04:22 PM	22.70	Normal
4	Nov 22, 2025 04:22 PM	22.52	Normal
5	Nov 17, 2025 04:22 PM	22.68	Normal
6	Nov 12, 2025 04:22 PM	22.43	Normal
7	Nov 07, 2025 04:22 PM	22.58	Normal
8	Nov 02, 2025 04:22 PM	22.35	Normal

ii. Pulse History

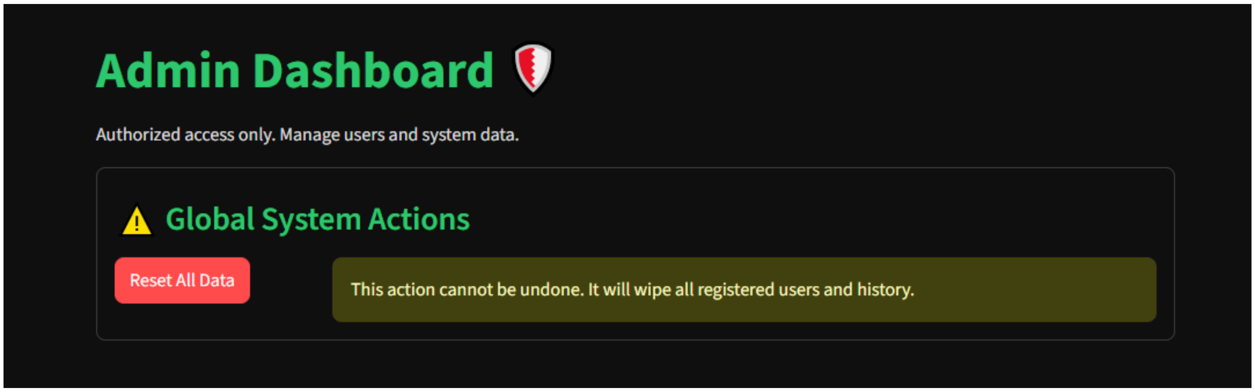


📊 Detailed History

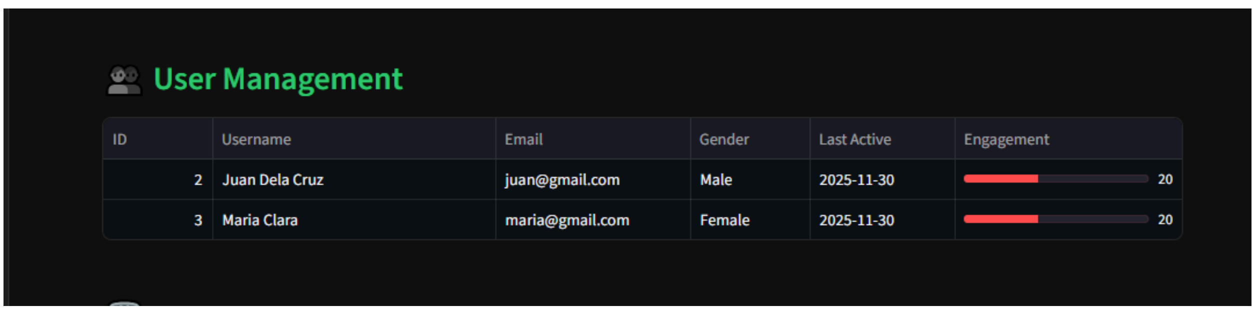
	Date Recorded	Heart Rate	Status
0	Dec 02, 2025 04:40 PM	60 bpm	Normal
1	Dec 02, 2025 04:39 PM	60 bpm	Normal
2	Dec 01, 2025 04:22 PM	76 bpm	Normal
3	Nov 30, 2025 04:22 PM	64 bpm	Normal
4	Nov 29, 2025 04:22 PM	69 bpm	Normal
5	Nov 28, 2025 04:22 PM	64 bpm	Normal
6	Nov 27, 2025 04:22 PM	71 bpm	Normal
7	Nov 26, 2025 04:22 PM	62 bpm	Normal
8	Nov 25, 2025 04:22 PM	69 bpm	Normal
9	Nov 24, 2025 04:22 PM	80 bpm	Normal

=== ADMIN ACCOUNT ===

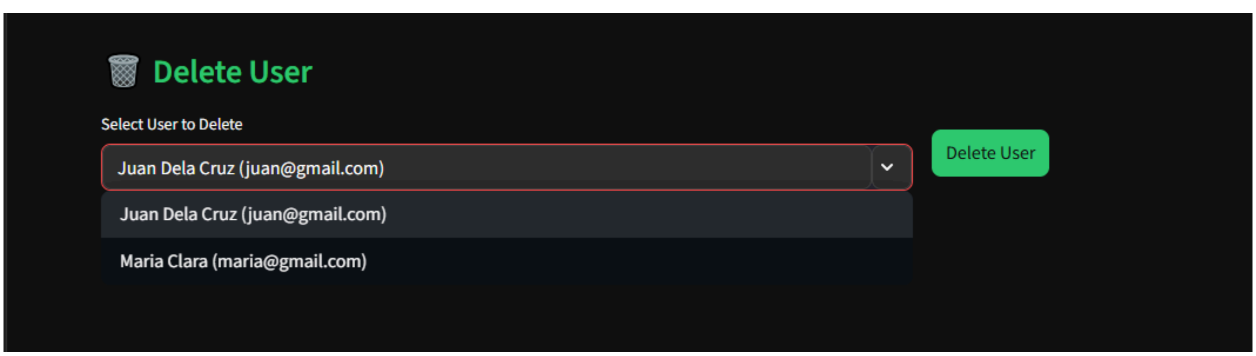
A. Global System Actions



B. User Management



C. Delete User



IV. MEMBERS AND THEIR ROLES

A. Sañoza, Rafael – Project Manager

- Responsible for the overall supervision of the project lifecycle. He coordinates team activities, manages project resources, and ensures that the system development adheres to the proposed timeline and meets all specific objectives. He acts as the primary liaison between the development team and stakeholders.

B. Toledana, Jan Louis – Lead Developer

- Tasked with the core technical implementation of the application. He handles the backend architecture, writes the Python source code, manages the database connectivity, and ensures the functionality of critical modules such as the BMI calculator and workout logging system.

C. Dizon, Bryan – UI/UX Designer

- In charge of the system's visual architecture and user experience. He designs the interface layouts, dashboard aesthetics, and navigation elements to ensure the application is intuitive, visually consistent, and user-friendly for both administrators and standard users.

D. **Antonio, Paul Zabdiel – Quality Assurance (QA) Specialist**

- Responsible for system testing and validation. He conducts rigorous testing protocols to identify functional errors or bugs, verifies the accuracy of biometric calculations, and ensures the system remains stable and secure under various usage conditions.